

Interarrival Time (IAT) Calculator

UAT Test Cases & Results

User Acceptance Testing Log · Portfolio Project

1. What This Document Is

Before trusting any calculator with a business decision, its numbers need to be checked against a known-correct answer. This document records how that was done for the IAT Calculator: a set of test cases was worked out by hand first, then the tool's actual output was compared against those hand-worked answers, row by row. Any mismatch was treated as a defect, root-caused, and re-tested until it passed.

This mirrors how User Acceptance Testing (UAT) works in a real business rollout - the tool doesn't get signed off until its output can be trusted against a known answer, not just because it "looks right."

2. Test Approach

- A small, hand-designed dataset was built specifically so every branch of the calculation logic would be exercised at least once (real receipt history, a new item with no history, two different fallback rules, and a case with no data anywhere).
- The correct answer for every test case was calculated manually before the tool was ever run, so there was no risk of the answer key being influenced by the tool's own output.
- Every result below was verified against that independent answer key.

3. Test Cases - IAT Calculation

16 SKU-Site combinations were tested, covering both site types (Hub and Spoke) and every fallback rule the system supports.

SKU	Site	Type	What This Case Tests	Expected (days)	Actual (days)
SKU1001	EU_HUB_01	Hub	Direct calculation - Hub formula (weekly)	14.0	14.0
SKU1002	EU_HUB_01	Hub	Direct calculation - Hub formula (weekly)	26.0	26.0
SKU1003	EU_SPOKE_01	Spoke	Direct calculation - Spoke formula (daily)	7.0	7.0
SKU2001	NA_HUB_01	Hub	Direct calculation - Hub formula (weekly)	30.3	30.3
SKU2002	NA_SPOKE_01	Spoke	Direct calculation - Spoke formula (daily)	10.7	10.7
SKU1050	EU_HUB_01	Hub	New SKU - resolved via GTIN match (same barcode family as SKU1001)	14.0	14.0
SKU1060	EU_HUB_01	Hub	New SKU - resolved via product hierarchy match (avg of SKU1001 + SKU1002)	20.0	20.0
SKU9999	EU_HUB_01	Hub	New SKU - no GTIN or hierarchy match found - falls back to default	182.0	182.0
SKU1001	EU_SPOKE_01	Spoke	Same SKU, different site - no receipt history at this specific site - default	182.0	182.0
SKU1002	EU_SPOKE_01	Spoke	Same pattern - no receipt history at this site - default	182.0	182.0
SKU1003	EU_HUB_01	Hub	Same pattern - no receipt history at this site - default	182.0	182.0
SKU2001	NA_SPOKE_01	Spoke	Same pattern - no receipt history at this site - default	182.0	182.0
SKU2002	NA_HUB_01	Hub	Same pattern - no receipt history at this site - default	182.0	182.0

SKU	Site	Type	What This Case Tests	Expected (days)	Actual (days)
SKU1050	EU_SPOKE_01	Spoke	No GTIN/hierarchy sibling with real data at this site - default	182.0	182.0
SKU1060	EU_SPOKE_01	Spoke	No GTIN/hierarchy sibling with real data at this site - default	182.0	182.0
SKU9999	EU_SPOKE_01	Spoke	No data anywhere for this SKU - default	182.0	182.0

4. Test Cases - Business Impact (Cycle Stock)

7 SKU-Site combinations with real demand and cost data were tested to confirm the tool correctly translates an IAT number into a business consequence (extra inventory needed, or inventory that can be released), including one deliberate control case where no gap should be found.

SKU	Site	Actual IAT	Naive Assumption	What This Case Tests	Expected Result
SKU1001	EU_HUB_01	14.0	7	Real delivery pattern is slower than assumed → buffer was too small	Understated - S
SKU1002	EU_HUB_01	26.0	30	Real delivery pattern is faster than assumed → buffer was too large	Overstated - Ex Cost
SKU1003	EU_SPOKE_01	7.0	7	Assumption matched reality exactly → control case, no error either way	No Gap
SKU2001	NA_HUB_01	30.3	14	Real delivery pattern is much slower than assumed → largest stockout risk in the set	Understated - S
SKU2002	NA_SPOKE_01	10.7	14	Real delivery pattern is faster than assumed → excess buffer tying up cash	Overstated - Ex Cost
SKU1050	EU_HUB_01	14.0	7	New SKU inheriting a sibling's real pattern, which differs from the naive assumption	Understated - S
SKU1060	EU_HUB_01	20.0	14	New SKU inheriting an averaged sibling pattern, which differs from the naive assumption	Understated - S

5. Defects Found & Resolved

Eight issues were found during testing and fixed before sign-off. They're listed here deliberately, not hidden - catching and fixing these is a core part of the actual work.

#	Severity	What Was Wrong	Root Cause	Resolution
1	Critical	SKU and Site values were swapped in the results table for several rows	Two separate calculation code paths existed and disagreed with each other; one read the source columns in the wrong order	Consolidated into a single calculation function used everywhere
2	Critical	Hub sites were calculated using the Spoke (daily) formula instead of the Hub (weekly) formula	The formula selection did not branch on site type - it defaulted to one formula for every site	Added explicit branching on site type before choosing a formula; added a regression check comparing both site types
3	High	SKUs were being evaluated at sites in a different region than their own (e.g. a Europe SKU tested at a North America site)	No region-matching rule existed when building the SKU x Site combinations to evaluate	Added a rule so a SKU is only evaluated at sites sharing its own region

#	Severity	What Was Wrong	Root Cause	Resolution
4	High	A new SKU's substitute value was calculated using another SKU's already-substituted (non-real) value, compounding inaccuracy	The substitution logic did not check whether a sibling's own value came from real data or from another substitution	Restricted substitution to only use siblings with a directly-calculated (real) value
5	Critical	The Business Impact tab showed demand, cost, and lead-time figures that did not match the uploaded source file at all	The uploaded file was not being read correctly by the app; it fell back to generated placeholder values	Rebuilt the data-read step to use only literal parsed contents of the uploaded file, with no fallback values allowed
6	Critical	Three SKUs showed the opposite risk conclusion of what the underlying numbers actually supported (e.g. labeled “excess buffer” when the data showed a shortfall)	Same root cause as Defect 5 - built on incorrect input data	Resolved automatically once Defect 5 was fixed; re-verified all rows against the hand-built answer key
7	Medium	The “Update Demand & Cost CSV” button did not trigger a re-upload or recalculation	The button was not connected to any file-picker or refresh action	Wired the button to a working file picker that replaces the dataset and recalculates immediately
8	Low	Dashboard used a decorative gradient banner, excessive icons, and non-semantic colors, inconsistent with a professional analytics tool	Initial build prioritized visual flair over data-tool conventions	Redesigned using a plain header, muted color palette, and color reserved only for risk status (red/amber/green)

6. Sign-Off Summary

Metric	Result
IAT calculation test cases	16 / 16 passed
Business impact test cases	7 / 7 passed
Defects logged	8
Defects resolved and re-tested	8
Open defects	0

All test cases passed after the fixes above were applied. The calculation logic, substitution rules, and business impact figures are considered verified for this dataset.